Lecture 14

Partial Derivatives

Example:

$$f(x,y) = \begin{cases} \dfrac{-xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 \quad , & (x,y) = (0,0) \end{cases}$$

Show that $f_x(x,y)$ and $f_y(x,y)$ exist at all points (x,y).

Solution: The partial derivatives of f are,

$$f_x(x,y) = \frac{\partial}{\partial x}\left(\frac{-xy}{x^2+y^2}\right)$$

$$= -\left[\frac{(x^2+y^2)y - xy(2x)}{(x^2+y^2)^2}\right]$$

$$= \frac{-x^2y - y^3 + 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{x^2y - y^3}{(x^2+y^2)^2}$$

Now, $f_y(x,y) = \frac{\partial}{\partial y}\left(\frac{-xy}{x^2+y^2}\right)$

$$= -\left[\frac{(x^2+y^2)x - xy(2y)}{(x^2+y^2)^2}\right]$$

$$= \frac{-x^3 - xy^2 + 2xy^2}{(x^2+y^2)^2}$$

$$= \frac{xy^2 - x^3}{(x^2+y^2)^2}$$

At point (0,0),

$$f_x(0,0) = \lim_{\Delta x \to 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\dfrac{-\Delta x \cdot 0}{(\Delta x)^2 + 0} - 0}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{0-0}{\Delta x} = 0$$

Similarly,

$$f_y(0,0) = \lim_{\Delta y \to 0} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \to 0} \frac{f(0,\Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \to 0} \frac{\frac{-0 \cdot \Delta y}{0 + (\Delta y)^2} - 0}{\Delta y}$$

$$= \lim_{\Delta y \to 0} \frac{0-0}{\Delta y} = 0$$

Therefore, f has partial derivatives at $(0,0)$ such that $f_x(0,0)$ and $f_y(0,0)$ exist. So, $f_x(x,y)$ and $f_y(x,y)$ exist at all points (x,y).

H.W.

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 \quad , & (x,y) = (0,0) \end{cases}$$

Show that $f_x(0,0)$ and $f_y(0,0)$ both exist.

Implicit Partial Differentiation

Example: Given $x^2 + y^2 + z^2 = 9$. Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

Solution: Given, $x^2 + y^2 + z^2 = 9$

So, $\frac{\partial}{\partial x}(x^2 + y^2 + z^2) = \frac{\partial}{\partial x}(9)$

$\Rightarrow 2x + 0 + 2z\frac{\partial z}{\partial x} = 0$

$\Rightarrow 2z\frac{\partial z}{\partial x} = -2x$

$\Rightarrow \frac{\partial z}{\partial x} = -\frac{2x}{2z} = -\frac{x}{z}$

Again, $x^2 + y^2 + z^2 = 9$.

So, $\frac{\partial}{\partial y}(x^2 + y^2 + z^2) = \frac{\partial}{\partial y}(9)$

$\Rightarrow 0 + 2y + 2z\dfrac{\partial z}{\partial y} = 0$

$\Rightarrow 2z\dfrac{\partial z}{\partial y} = -2y$

$\Rightarrow \dfrac{\partial z}{\partial y} = -\dfrac{2y}{2z} = -\dfrac{y}{z}$

<u>Note:</u> We performed implicit differentiation on the 3rd term z because $z = f(x, y)$, i.e., z is a dependent variable. If z was an independent variable, we would have taken $\frac{\partial(z^2)}{\partial x} = 0$ or $\frac{\partial(z^2)}{\partial y} = 0$ instead.

<u>Higher Order Partial Derivatives</u>

<u>Second Order Partial Derivatives:</u>

$f_{xx} = \dfrac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \dfrac{\partial^2 f}{\partial y^2}$

$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\, \partial x}$ (first differentiate with respect to x, then y)

$f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\, \partial y}$ (first differentiate with respect to y, then x)

- $f_{xy} = f_{yx}$

<u>Third Order Partial Derivatives:</u>

$f_{xxx} = \dfrac{\partial^3 f}{\partial x^3}, \qquad f_{yyy} = \dfrac{\partial^3 f}{\partial y^3}$

Mixed third-order partial derivatives:

$f_{xxy},\ f_{xyx},\ f_{yxx},\ f_{yyx}$ etc.

<u>Example:</u> Let $f(x, y) = e^{x^2+xy+y^2}$. Find $f_{xx}, f_{yy}, f_{xy}, f_{yx}$. Show that $f_{xy} = f_{yx}$.

<u>Solution:</u> Given, $f(x, y) = e^{x^2+xy+y^2}$

So, $f_x = \frac{\partial}{\partial x}\left(e^{x^2+xy+y^2}\right) = e^{x^2+xy+y^2}.(2x + y)$

$f_y = \dfrac{\partial}{\partial y}\left(e^{x^2+xy+y^2}\right) = e^{x^2+xy+y^2}.(x + 2y)$

Now,

$$f_{xx} = \frac{\partial}{\partial x}(f_x) = \frac{\partial}{\partial x}\{e^{x^2+xy+y^2}(2x+y)\}$$

$$= e^{x^2+xy+y^2}.(2) + e^{x^2+xy+y^2}(2x+y)(2x+y)$$

$$= e^{x^2+xy+y^2}\{2+(2x+y)^2\}$$

$$f_{yy} = \frac{\partial}{\partial y}(f_y) = \frac{\partial}{\partial y}\{e^{x^2+xy+y^2}(x+2y)\}$$

$$= e^{x^2+xy+y^2}.(2) + e^{x^2+xy+y^2}(x+2y)(x+2y)$$

$$= e^{x^2+xy+y^2}\{2+(x+2y)^2\}$$

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = \frac{\partial}{\partial y}\{e^{x^2+xy+y^2}(2x+y)\}$$

$$= e^{x^2+xy+y^2}.(1) + e^{x^2+xy+y^2}(x+2y)(2x+y)$$

$$= e^{x^2+xy+y^2} + e^{x^2+xy+y^2}(x+2y)(2x+y)$$

$$f_{yx} = \frac{\partial}{\partial x}(f_y) = \frac{\partial}{\partial x}\{e^{x^2+xy+y^2}(x+2y)\}$$

$$= e^{x^2+xy+y^2}.(1) + e^{x^2+xy+y^2}(2x+y)(x+2y)$$

$$= e^{x^2+xy+y^2} + e^{x^2+xy+y^2}(2x+y)(x+2y)$$

Here, we can see that, $f_{xy} = f_{yx}$.

<u>Example:</u> Find the rate of change of z with respect to x and y at the points (3,1) and $(-2,4)$, for $z = \sin(y^2 - 4x)$.

<u>Solution:</u> Here, $z = \sin(y^2 - 4x)$. So, $z = f(x, y) = \sin(y^2 - 4x)$.

The rate of change of z with respect to x,

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\{\sin(y^2 - 4x)\} = \cos(y^2 - 4x).(-4) = -4\cos(y^2 - 4x)$$

The rate of change of z with respect to y,

$$\frac{\partial z}{\partial \mathrm{y}} = \frac{\partial}{\partial y}\{\sin(y^2 - 4x)\} = \cos(y^2 - 4x).(2y) = 2y\cos(y^2 - 4x)$$

At point (2,1):

$$\frac{\partial z}{\partial x}|_{(2,1)} = -4\cos(1^2 - 4 \cdot 2) = -4\cos(-7)$$

$$\frac{\partial z}{\partial y}|_{(2,1)} = 2 \cdot 1\cos(1^2 - 4 \cdot 2) = 2\cos(-7)$$

At point $(-2,4)$:

$$\frac{\partial z}{\partial x}|_{(-2,4)} = -4\cos\left(4^2 - 4(-2)\right) = -4\cos(16+8) = -4\cos(24)$$

$$\frac{\partial z}{\partial y}|_{(-2,4)} = 2 \cdot 4\cos\left(4^2 - 4(-2)\right) = 8\cos(16+8) = 8\cos(24)$$

<u>Partial Derivatives of Functions with More Than Two Variables</u>

$w = f(x, y, z)$.

<u>First Order Partial Derivatives:</u>

$$\frac{\partial w}{\partial x} = f_x(x, y, z) = \frac{\partial f}{\partial x}$$

$$\frac{\partial w}{\partial y} = f_y(x, y, z) = \frac{\partial f}{\partial y}$$

$$\frac{\partial w}{\partial z} = f_z(x, y, z) = \frac{\partial f}{\partial z}$$

<u>Second Order Partial Derivatives:</u>

$$\frac{\partial^2 f}{\partial x^2} = f_{xx}, \frac{\partial^2 f}{\partial y^2} = f_{yy}, \frac{\partial^2 f}{\partial z^2} = f_{zz}, \frac{\partial^2 f}{\partial x\, \partial y} = f_{xy}, \frac{\partial^2 f}{\partial x\, \partial z} = f_{xz}, \frac{\partial^2 f}{\partial y\, \partial z} = f_{yz}$$

<u>Third Order Partial Derivatives:</u>

$$f_{xxx} = \frac{\partial^3 f}{\partial x^3}, \quad f_{yyy} = \frac{\partial^3 f}{\partial y^3}, \quad f_{zzz} = \frac{\partial^3 f}{\partial z^3}$$

$$f_{xyz} = \frac{\partial^3 f}{\partial z\, \partial y\, \partial x} = \frac{\partial^3 f}{\partial x\, \partial y\, \partial z} = f_{zyx} = f_{zxy} = f_{xzy} = f_{yxz} = f_{yzx}$$

$f_{xxy} = \frac{\partial^3 f}{\partial y\, \partial x\, \partial x}$ etc.

<u>nth Order Partial Derivatives:</u>

$\frac{\partial^n f}{\partial x^n}, \frac{\partial^n f}{\partial y^n}, \frac{\partial^n f}{\partial z^n}$ … etc.

<u>Example:</u> Find $f_{xy}, f_{yz}, f_{xz}, f_{zz}, f_{zyy}, f_{zxy}, f_{zyx}, f_{xxyz}$ for $f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z$.

<u>Solution:</u> Given, $f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z$.

Now, $f_x = \frac{\partial}{\partial x}(x^3y^5z^7 + xy^2 + y^3z) = 3x^2y^5z^7 + y^2$

$$f_y = \frac{\partial}{\partial y}(x^3y^5z^7 + xy^2 + y^3z) = 5x^3y^4z^7 + 2xy + 3y^2z$$

$$f_z = \frac{\partial}{\partial z}(x^3y^5z^7 + xy^2 + y^3z) = 7x^3y^5z^6 + y^3$$

Now,

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = 15x^2y^4z^7 + 2y = f_{yx}$$

$$f_{yz} = \frac{\partial}{\partial z}(f_y) = 35x^3y^4z^6 + 3y^2 = f_{zy}$$

$$f_{xz} = \frac{\partial}{\partial z}(f_x) = 21x^2y^5z^6 = f_{zx}$$

$$f_{zz} = \frac{\partial}{\partial z}(f_z) = 42x^3y^5z^5$$

Now,

$$f_{zxy} = \frac{\partial^3 f}{\partial y\, \partial x\, \partial z}$$

$$= \frac{\partial^2}{\partial y\, \partial x}\left(\frac{\partial f}{\partial z}\right)$$

$$= \frac{\partial^2}{\partial y\, \partial x}(7x^3y^5z^6 + y^3)$$

$$= \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}(7x^3y^5z^6 + y^3)\right)$$

$$= \frac{\partial}{\partial y}(21x^2y^5z^6)$$

$$= 105x^2y^4z^6$$

We know that, $f_{zxy} = f_{zyx}$.

Hence, $f_{zyx} = 105x^2y^4z^6$

Now,

$$f_{zyy} = \frac{\partial^3 f}{\partial y\, \partial y\, \partial z} = \frac{\partial^2}{\partial y^2}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial^2}{\partial y^2}(7x^3y^5z^6 + y^3)$$

$$= \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}(7x^3y^5z^6 + y^3)\right)$$

$$= \frac{\partial}{\partial y}(35x^3y^4z^6 + 3y^2)$$

$= 140x^3y^3z^6 + 6y$

Now,

$$f_{xxyz} = \frac{\partial^4 f}{\partial z\, \partial y\, \partial x\, \partial x} = \frac{\partial^3}{\partial z\, \partial y\, \partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial^3}{\partial z\, \partial y\, \partial x}(3x^2y^5z^7 + y^2)$$

$$= \frac{\partial^2}{\partial z\, \partial y}\left(\frac{\partial}{\partial x}(3x^2y^5z^7 + y^2)\right)$$

$$= \frac{\partial^2}{\partial z\, \partial y}(6xy^5z^7 + 0)$$

$$= \frac{\partial}{\partial z}\left(\frac{\partial}{\partial y}(6xy^5z^7)\right)$$

$$= \frac{\partial}{\partial z}(30xy^4z^7)$$

$$= 210xy^4z^6$$

<u>Example:</u> Let, $f(x, y, z) = x^6 + xy^2 + z^3y$. Find first and second order partial derivatives.

<u>Solution:</u> Given, $f(x, y, z) = x^6 + xy^2 + z^3y$

<u>First Order Partial Derivatives</u>

$f_x(x, y, z) = 6x^5 + y^2$

$f_y(x, y, z) = 2xy + z^3$

$f_z(x, y, z) = 3z^2y$

<u>Second Order Partial Derivatives</u>

$f_{xx}(x, y, z) = 30x^4$

$f_{yy}(x, y, z) = 2x$

$f_{zz}(x, y, z) = 6zy$

$f_{xy}(x, y, z) = 2y$

$f_{yx}(x, y, z) = 2y$

$f_{xz}(x, y, z) = 0$

$f_{zx}(x, y, z) = 0$

$f_{yz}(x, y, z) = 3z^2$

$f_{zy}(x, y, z) = 3z^2$

<u>Example:</u> Let, $f(x, y, z) = x^3 + y^3 + z^3$. Show that,

$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 3(x^2 + y^2 + z^2)$ and $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 6(x + y + z)$.

<u>Solution:</u>

First derivatives,

$$\frac{\partial f}{\partial x} = 3x^2, \qquad \frac{\partial f}{\partial y} = 3y^2, \qquad \frac{\partial f}{\partial z} = 3z^2$$

So, $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$. [Showed]

Second derivatives,

$$\frac{\partial^2 f}{\partial x^2} = 6x, \qquad \frac{\partial^2 f}{\partial y^2} = 6y, \qquad \frac{\partial^2 f}{\partial z^2} = 6z$$

So, $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 6x + 6y + 6z = 6(x + y + z)$. [Showed]

<u>H.W. Practice Sheet</u>

<u>H.W. 13.3 (37–45)</u>